MagicCode Super-Resolution Software API Reference

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Header: interface/mc_interface.h. Public MC_ERROR_* codes are in interface/mc_error.h (included by mc_interface.h). v3.0.0 keeps the nscaler API, takes two input frames on mc_nscaler_enable, and adds GEN/FG. v2.2.0 removed mc_enable.h and the MC_Enable helper.

1. Overview

The native API is a C interface for initialization, processing, control, status query, release, and version query. C++ projects include the header normally; symbols use extern "C".

2. Core Workflow

1. Zero-init ctrl_param_t and set input_type, model_path, scaler_factor, alg_mode, backend, and (for graphic) gpu_context plus create-stage depth/HDR. Image size is per frame on mc_nscaler_input_frame_t, not on ctrl_param_t.
2. Create with mc_nscaler_control(&handle, MC_NSCALER_CMD_SET_PARAM, &ctrl, NULL), or let the first mc_nscaler_enable create with defaults (2x SR_SPEED_MODE_FOR_IMAGE). Keep the opaque handle.
3. Each frame, fill mc_nscaler_input_frame_t (color + width/height) and mc_nscaler_output_frame_t. Image may leave frame NULL. Graphic must set frame to a temporal_frame_t with struct_size, depth, motion, jitter, and camera fields.
4. Call mc_nscaler_enable(&handle, in2, &out_frame) with mc_nscaler_input_frame_t *in2[2] = { &in_frame, NULL } (set in2[1] only for GEN interpolate).
5. Optionally call mc_nscaler_control with MC_NSCALER_CMD_SET_PARAM or MC_NSCALER_CMD_QUERY_STATUS.
6. Call mc_nscaler_disable exactly once for each successfully created handle.

3. Enumerations

Enum	Values	Notes
mc_cmd_e	MC_NSCALER_CMD_SET_PARAM, MC_NSCALER_CMD_QUERY_STATUS	SET_PARAM/QUERY_STATUS use ctrl_param_t / output_status_params_t.
input_type_e	INPUT_BUFFER / R8, INPUT_TEXTURE_RGB8Unorm, INPUT_TEXTURE_R8Unorm, INPUT_BUFFER_RGB	CPU buffers vs GPU textures.
alg_mode_e	SR_SPEED_MODE_FOR_IMAGE, SR_BALANCED_MODE_FOR_IMAGE,	SR 0..3; GEN 4 or 8 (not both). Combinable (example 2|4=6).

	SR_SPEED_MODE_FOR_GRAPHIC, SR_BALANCED_MODE_FOR_GRAPHIC, GEN_SPEED_MODE_FOR_GRAPHIC, GEN_BALANCED_MODE_FOR_GRAPHIC	MAX_ALG_MODE=16. Public: MAX_ALG_MODE and MC_ALG_GEN_MASK.
magic_backend_e	DEFAULT, X86, NEON, METAL, OPENGL, OPENGLES, VULKAN, D3D11	Runtime selector.
mc_mv_jitter_e	UNSPECIFIED/EXCLUDED/INCLUDED	On temporal_frame_t.mv_jitter.
mc_depth_reversed_e / infinite / hdr	UNSPECIFIED plus conventional/reversed, finite/infinite, LDR/HDR	Create-stage on ctrl_param_t.

4. Structures

4.1 ctrl_param_t

- No struct_size and no width/height: ctrl_param_t is versioned by MC_CTRL_PARAM_ABI_VERSION; the image size lives on the frame structs.
- scaler_factor image [1, 8], graphic (1, 8]. Frame width/height [64, 4032] on mc_nscaler_input_frame_t.
- alg_mode: SR bits 0..3 plus optional GEN_SPEED=4 or GEN_BALANCED=8 (not both). MAX_ALG_MODE=16. See the mode table.
- gpu_context: copied at create; immutable. Vulkan graphic requires physical_device + device. D3D11 graphic requires device. Metal NULL device = system default.
- depth_reversed / depth_infinite / hdr_color: graphic create-stage; ignored for image. Immutable after create together with model_path, gpu_context, and backend; SET_PARAM ignores changes to them.

4.2 mc_nscaler_input_frame_t / mc_nscaler_output_frame_t and temporal_frame_t

- mc_nscaler_input_frame_t holds the color resource plus per-frame width/height; mc_nscaler_output_frame_t holds the output resource. Both use the magic_resource_t handle/format/layout/target/mip_count layout.
- mc_nscaler_input_frame_t.frame is NULL for image and required for graphic.
- mc_nscaler_input_frame_t.command_buffer: Vulkan graphic recording VkCommandBuffer; optional MTLCommandBuffer on Metal; unused on D3D11/GL/GLES.
- temporal_frame_t.struct_size must cover the v2 prefix through transparency. Tail fields (mv_jitter, exposure_texture) are read only when struct_size covers them.
- Required graphic resources: depth, motion. Optional: reactive, transparency, exposure_texture (1×1 R32F).

4.3 magic_resource_t

handle is a union: pointer (CPU / MTLTexture* / ID3D11Texture2D*), vk_image (VkImage), or gl_texture (GLuint). Exactly one member is live, chosen by the create-time backend. format is backend-native;

layout is VkImageLayout (zero for other backends). No width/height: matching handle sizes is a caller contract.

4.4 ctrl_param_t / output_status_params_t

SET_PARAM updates scale and mode on a live handle and ignores immutable fields (model_path, gpu_context, backend, depth_reversed/infinite/hdr_color); size changes arrive through mc_nscaler_enable frame sizes. QUERY_STATUS returns dimensions, scale, mode, backend, gpu_time, error_code, and when temporal_status_valid is set, the last successful graphic snapshot (scalars only). GPU handles and command_buffer are never stored.

5. Functions

Function	Purpose	Return
int mc_nscaler_enable(void**, mc_nscaler_input_frame_t *in_frame[2], mc_nscaler_output_frame_t*)	Create (*handle NULL) and/or process. in_frame[0] required; in_frame[1] only for GEN interpolate	0 or negative MC_ERROR_*
int mc_nscaler_control(void**, mc_cmd_e, ctrl_param_t*, output_status_params_t*)	SET_PARAM or QUERY_STATUS	0 or negative
int mc_nscaler_disable(void*)	Release handle	0 or negative
char* mc_nscaler_version(void)	Version string (static in this module; copy before unload)	Static "v3.0.0"; never NULL
const char* mc_build_id(void)	Compile stamp (version + date + time). Same lifetime as mc_nscaler_version	Never NULL; invalid after unload
void mc_loaded_module_path(char*, int)	UTF-8 path of the loaded module (Windows GetModuleFileName). Empty when unavailable	Writes buf

5.1 Vulkan graphic lifecycle

Keep the handle across frames. mc_nscaler_enable only records into the caller-owned recording VkCommandBuffer. Set input color layout, graphic depth/motion.layout, and output layout to the real VkImageLayout at entry (0 is UNDEFINED and fails). The library does not begin, end, submit, or wait. After the caller submits and waits, call mc_nscaler_disable at shutdown — not immediately after mc_nscaler_enable while GPU work may still be in flight.

5.2 Frame generation (GEN)

- Create with a GEN bit in ctrl_param_t.alg_mode. Vulkan + MAGIC_SR_HAS_FG only (Windows/Linux product builds). Android Vulkan currently has no MAGIC_SR_HAS_FG. Other backends/builds return MC_ERROR_INIT_ALG_MODE_GEN_NOT_AVAILABLE (-100050) or MC_ERROR_CONTROL_ALG_MODE_GEN_NOT_AVAILABLE (-102013).
- in_frame[1]==NULL: existing SR path on in_frame[0]. in_frame[1]!=NULL: interpolate t=0.5 between [0] and [1] into out_frame.

- GEN color on [0]/[1]/out is output (post-SR) resolution. Depth and motion stay at render (input) resolution; FG samples with normalized UV.
- Both frames need color, depth, and temporal_frame_t. At least one motion texture is required. Record into in_frame[0].command_buffer; NULL returns MC_ERROR_TEMPORAL_COMMAND_BUFFER (-105017). Caller submits. Output layout after return is VK_IMAGE_LAYOUT_GENERAL.
- in_frame[0].width/height must equal the handle render input size. A different non-zero size recreates the handle (same as SR resize).
- GEN_SPEED → FG-Fast (MC_CreateFrameGenFast internally; no weights). GEN_BALANCED → FG-CoreV3 (ctrl_param_t.model_path is the FG weight file; empty uses default search).

5.3 GEN_BALANCED weights (model_path)

- Set ctrl_param_t.model_path on the creating SET_PARAM to the FG weight file.
- Empty model_path uses default search: MAGIC_FG_WEIGHT_PATH, then magic_fg_core.bin, then model/magic_fg_core.bin. Packaged file model/magic_fg_core.bin is 331,136 bytes.
- model_path is immutable after create. To change weights, call mc_nscaler_disable and create a new handle.
- MC_ERROR_INIT_FG_WEIGHTS / MC_ERROR_CONTROL_FG_WEIGHTS (-100051 / -102014); MC_ERROR_INIT_FG_CREATE_FAILED / MC_ERROR_CONTROL_FG_CREATE_FAILED (-100052 / -102015).

5.4 Version string lifetime and exported symbols

- mc_nscaler_version() returns the static MC_VERSION string in this module; never NULL. The pointer is invalid after FreeLibrary/dlclose — copy first. That misuse was the root cause of the previous DLL-unload crash.
- mc_build_id() is a compile stamp (version, __DATE__, __TIME__); same lifetime.
- Windows DLL exports only: mc_nscaler_control, mc_nscaler_disable, mc_nscaler_enable, mc_nscaler_version, mc_build_id, mc_loaded_module_path.

6. Removed APIs

v2.2.0 removed the v2.0.0 session entry (MC_Init / MC_Process / MC_Uninit / MC_Control) and the v2.1.0 helper (MC_Enable / MC_Disable / MC_Enable_SetModelPath / MC_Enable_SetInputSizeHint) together with mc_enable.h. Use mc_nscaler_enable / mc_nscaler_control / mc_nscaler_disable / mc_nscaler_version. A NULL *handle on mc_nscaler_enable creates a default 2x SR_SPEED_MODE_FOR_IMAGE session.

Graphic Super-Resolution

Graphic super-resolution runs through the nscaler session API: libmagic_sr.a (Windows: libmagic_sr.lib) + mc_interface.h + mc_nscaler_enable / mc_nscaler_control / mc_nscaler_disable. v2.2.0 replaced the v2.0.0 MC_Init / MC_Process / MC_Uninit entry and removed the v2.1.0 MC_Enable helper (mc_enable.h is deleted). A NULL *handle on mc_nscaler_enable creates a session with defaults (2x

SR_SPEED_MODE_FOR_IMAGE) and processes the first frame. Pass mc_nscaler_input_frame_t *in2[2] = { &in, NULL } (set in2[1] only for GEN interpolate).

Modes

Value	Enumerator	Role
0	SR_SPEED_MODE_FOR_IMAGE	Image, throughput first
1	SR_BALANCED_MODE_FOR_IMAGE	Image, quality/cost trade-off
2	SR_SPEED_MODE_FOR_GRAPHIC	Graphic, throughput first
3	SR_BALANCED_MODE_FOR_GRAPHIC	Graphic, canonical FSR-family path
4	GEN_SPEED_MODE_FOR_GRAPHIC	FG-Fast (analytic, no weights). Vulkan + MAGIC_SR_HAS_FG. Combinable with SR bits; bare 4 equals SR_SPEED_MODE_FOR_IMAGE|GEN_SPEED.
8	GEN_BALANCED_MODE_FOR_GRAPHIC	FG-CoreV3. Same availability as Fast. Needs magic_fg_core.bin. Do not set both GEN bits.

- Graphic scaler_factor is (1, 8]; 1.0 itself is rejected. Image remains [1.0, 8.0].
- SR_SPEED_MODE_FOR_GRAPHIC on Metal, Vulkan, and OpenGLES currently uses a lightweight Convert → Activate → Upscale pipeline. Direct3D 11 and desktop OpenGL 4.3 Speed use the FSR-family graphic path (same family as Balanced, not a separate customer API).
- SR_BALANCED_MODE_FOR_GRAPHIC uses the canonical FSR-family path on every GPU graphic backend. Desktop Balanced edge-resolve is supported on Windows Vulkan, Windows Direct3D 11, desktop OpenGL 4.3, and Apple Metal. Android Vulkan and Android OpenGLES do not run edge-resolve. If enable_sharpening is on, RCAS runs after edge-resolve.
- CPU backends (X86 / NEON) cannot run graphic modes.
- The API accepts (1, 8]. Exact 2× has optimized kernels. Not every scale or device is claimed as device-run.
- Internal pipeline names are implementation detail, not public configuration.

Minimum per-frame inputs

- mc_nscaler_input_frame_t / mc_nscaler_output_frame_t: caller-owned color and output GPU resources.
- mc_nscaler_input_frame_t.frame: non-NULL temporal_frame_t* with struct_size = sizeof(temporal_frame_t).
- temporal_frame_t.depth and motion at input resolution.
- jitter_offset_x/y in input pixels, top-left, +X right, +Y down.
- frame_index / reset_history; camera_near / camera_far / camera_fov_y (radians).
- gpu_context at create (mc_nscaler_control SET_PARAM, or the first mc_nscaler_enable): Vulkan requires physical_device + device; D3D11 requires device; Metal device may be NULL; GL/GLES use the current context (library never makes current).
- Vulkan graphic requires mc_nscaler_input_frame_t.command_buffer to be a recording VkCommandBuffer. Metal command_buffer is optional. D3D11 / GL / GLES leave it NULL.

- Create-stage depth_reversed / depth_infinite / hdr_color on ctrl_param_t (all-zero = conventional-Z, finite far, LDR). Immutable for the handle lifetime.

Motion, depth, and masks

- Motion is current → previous. The library never negates MVs.
- motion_vector_scale_x/y is the only MV numeric conversion. (0,0) defaults to (input_width, input_height). Mixed zero is rejected.
- Depth is GPU device depth in [0, 1]. Linear view-space depth is not accepted.
- reactive / transparency are optional. Explicit caller textures win per channel.
- When omitted, the library derives an approximate mask. That is not a semantic transparency mask.
- Speed on Metal/Vulkan/GLES may derive both missing reactive and an approximate transparency. FSR-family paths (Balanced on all backends; Speed on D3D11/desktop GL) derive missing reactive only and bind an internal zero transparency texture.
- Both graphic modes enable depth+MV history rejection by default.
- Diagnostic only: MAGICSR_TAAU_AUTO_MASK=0 and MAGICSR_TAAU_HIST_REJECT=0 can turn those defaults off. They are not a customer configuration path.

7. Selected Graphic error codes (Selected temporal error codes). Full public list: interface/mc_error.h (included by mc_interface.h).

- MC_ERROR_TEMPORAL_NULL_FRAME (-105001)
- MC_ERROR_TEMPORAL_STRUCT_SIZE (-105002)
- MC_ERROR_TEMPORAL_MISSING_COLOR/OUTPUT/DEPTH/MOTION (-105003..-105006)
- MC_ERROR_TEMPORAL_NONFINITE / CAMERA_NEAR_FAR / MV_SCALE (-105007, -105008, -105011)
- MC_ERROR_TEMPORAL_FORMAT (-105015); Vulkan UNDEFINED layout is rejected
- MC_ERROR_TEMPORAL_COMMAND_BUFFER (-105017) Vulkan recording buffer required
- MC_WARNING_TEMPORAL_FRAME_INDEX (105001) not last+1
- MC_ERROR_INIT_ALG_MODE_GEN_NOT_AVAILABLE (-100050); MC_ERROR_CONTROL_ALG_MODE_GEN_NOT_AVAILABLE (-102013)
- MC_ERROR_INIT_FG_WEIGHTS (-100051); MC_ERROR_CONTROL_FG_WEIGHTS (-102014)
- MC_ERROR_INIT_FG_CREATE_FAILED (-100052); MC_ERROR_CONTROL_FG_CREATE_FAILED (-102015)
- Vulkan Speed (Lite-TAAU) now lets an explicit strong reactive mask suppress the still-camera accumulation branch. Previously that branch ignored the caller reactive mask when the camera was still or per-pixel MV was below about 0.2 px, so pixels with ~0 MV but changing content (TV / video-wall / LED-screen textures) could keep ghosting for about 64–96 frames after a content cut. Strong reactive (≥ about 0.85, fully effective at ≥ 0.95) now weakens that accumulation. Omitting reactive or passing a weak value (< 0.85) is unchanged. Recommendation: mark screen-like objects whose MV is 0 but whose content changes (for example with CustomDepth stencil) as reactive at strength 1.0. Limit: Vulkan only in this release.

The Metal / GLES / D3D11 Speed still-accumulation structure is the same idea but is not modified or verified here.